

WPP Enterprise Agent Framework — Solution Architecture

1. Approach

GHCP SDK agentic loops running on Azure AI Foundry Hosted Agents, orchestrated by Azure Durable Functions, governed by Fleet Manager agents, centrally managed through APIM AI Gateway and Foundry Control Plane.

Ref: [GHCP SDK](#) (MIT, Python/TypeScript/Go/.NET)

Agentic loop: an ephemeral GHCP SDK session triggered by an event. Loads skills and MCP tools for the task, does the work, writes state to a durable store, emits telemetry, and exits.

Ref: [Agent loop docs](#)

Skills: [SKILL.md files](#) define capabilities declaratively. New capability = new skill file. Skills are the specialisation mechanism — instead of separate agents for screening, sourcing, compliance, etc., a single domain-scoped Hosted Agent loads different skills per workflow phase. Each skill defines its own role, allowed tools, model assignment, and governance rules. Skills are also crystallisation artefacts — proven agentic patterns graduate from LLM-generated to deterministic code, versioned as skills. Skills are registered in Azure API Center with lifecycle management (Design → Preview → Production → Deprecated) and governed via APIM AI Gateway.

Ref: [Azure API Center lifecycle](#)

MCP tools: enterprise system integrations (Workday, Greenhouse, LinkedIn, ServiceNow, Graph, Dataverse) exposed as [MCP servers](#), governed through APIM AI Gateway with auth, rate limiting, and content safety policies. APIM provides a [REST-to-MCP gateway](#) that auto-generates MCP tool definitions from OpenAPI specs.

Hooks: GHCP SDK session hooks ([onPreToolUse](#) , [onPostToolUse](#)) provide operational governance inside the agentic loop. Non-revocable actions (send email, submit background check, extend offer, execute payment) are intercepted by hooks, blocked from immediate execution, and routed to the human's PA for approval. After approval, execution is deterministic — no LLM in the send path. Hooks also handle audit logging (every tool call logged with workflow context) and tool allow-listing per skill set.

Human interaction: humans interact with their personal agent (PA) through M365 Copilot (Teams, Outlook) via the [M365 Agents SDK](#), and through the custom Control Plane UI for fleet management. There is always an agentic layer between humans and the system.

2. Architecture

Three Tiers

Tier	Component	Role
1. Fleet Manager	Always-on GHCP SDK Hosted Agent	Consumes telemetry from all workflows. Reasons about fleet health, SLA risk, anomalies. Composes the exception queue for the Control Plane. Monitors compliance enforcement events.
2. Workflow Orchestration	Azure Durable Functions (GA)	One instance per workflow. Routes events to agentic loops. HITL waits at zero compute. Timer escalation. Parallel coordination. Checkpoint/replay.
3. Agentic Loops	Ephemeral GHCP SDK sessions	Triggered per-phase. Load skills + MCP tools. Reason about the task. Write state. Emit OTEL telemetry . Exit.

Central Governance

Layer	Technology	What It Manages
APIM AI Gateway	Azure API Management (GA)	Models: load balancing , failover , token limits , cost tracking , semantic caching . MCP Tools: auth, rate limiting, content safety , discovery . A2A Agents: agent card registration , JSON-RPC mediation , policy enforcement (preview). Skills: lifecycle management via Azure API Center , GitHub Actions integration, allowed tools governance.
Foundry Control Plane	Microsoft Foundry (GA)	Platform governance. Custom agents register via portal/API (requires APIM AI Gateway as proxy). Observability via Application Insights (run rates, error rates, token usage, cost, execution traces per agent). Agent fleet inventory . Model registry . Tool registry. Guardrails configuration . Built-in evaluators . Note: Foundry Control Plane provides platform-level monitoring. WPP's operator-specific requirements (exception-only surfacing, bulk HITL, autonomy dials, situational awareness, skill amplification) require the custom Control Plane UI. The custom UI consumes data from Application Insights APIs and Foundry REST APIs.

Layer	Technology	What It Manages
Agent 365	Microsoft Agent 365 (GA May 2026, currently preview)	Agent identity via Entra Agent ID . Agent lifecycle management (activate, block, delete) in M365 admin center with agent registry . Purview and Defender integration. Conditional Access for agents . Note: Agent 365 is not yet GA. The exact integration with Foundry Hosted Agents needs validation during POC — it is unclear whether Hosted Agents register automatically in Agent 365 or require manual onboarding. Agent 365 natively supports Copilot Studio agents; our custom Control Plane UI will also natively support Copilot Studio agents alongside GHCP SDK agents.

Identity & Access

The Entra Agent ID lives on the **Hosted Agent container**, not on individual sessions. The GHCP SDK session is a unit of work inside a persistent container — it uses the container's identity when calling tools.

Hosted Agent topology (design decision for POC):

Hosted Agent	Entra Agent ID	Tool Access	Runs Loops For
Hiring Agent	hiring-agent@wpp	Greenhouse, LinkedIn, Workday (hiring), Graph, ACS	All hiring workflow phases (screening, sourcing, compliance, offer, onboarding)
Finance Agent	finance-agent@wpp	Workday (finance), D365 F&O, Maconomy	All P2P workflow phases (intake, validation, routing, payment, reconciliation)
Fleet Manager	fleet-manager@wpp	Read-only: Foundry Tracing, Event Grid, workflow state store. No tool invocation on downstream systems.	Fleet monitoring, exception composition, compliance oversight

Each Hosted Agent is domain-scoped — it has only the tool access its domain requires. All loops running inside it inherit that scope.

OBO vs app-only: when a loop is triggered by a human action (Finance BP approves), it can act [on-behalf-of](#) that human — audit trail attributes the decision to them, downstream access uses their delegated permissions. For autonomous phases (screening 200 CVs), the loop uses the Hosted Agent's app-only identity.

Skill and policy governance: skill promotion (Design → Production) goes through Azure API Center lifecycle gates. API Center integrates with GitHub Actions workflows for syncing SKILL.md files from repositories. Autonomy threshold changes are audit-logged with operator identity. Governance policy changes require PR review in Git, deployed via [APIOps CI/CD](#).

Human Interaction Model

Every human has a **personal agent (PA)** — a GHCP SDK agent surfaced in M365 Copilot via M365 Agents SDK. The PA knows the human's role, permissions, and context. It surfaces information, recommends actions, drafts outputs, and triggers workflows on behalf of the human. Humans make all decisions. The PA prepares and executes.

Surface	How	Who
M365 Copilot (Teams)	Each user's PA agent. Understands role context, surfaces relevant information proactively, triggers workflows OBO the user when instructed.	All WPP users
Email (Adaptive Cards)	PA composes Adaptive Card with context and recommendation. Human reviews and decides. Response routes back through PA.	Finance BP, approvers
Control Plane UI	A richer surface the PA uses when the situation requires more than a Teams message — fleet view, bulk approvals, drill-downs. The HR BP's PA IS the Fleet Manager.	HR BP, Fleet operators
ServiceNow	PA writes provisioning tasks via ServiceNow MCP.	IT Ops
Web portal	Custom web surface for external participants.	Candidates
Voice	GPT-Realtime as speech-to-speech front end. Tool calls to GHCP SDK backend for reasoning.	Candidates (screening)

Flow (Hiring Workflow)

1. **Hiring Manager's PA** surfaces a headcount gap based on Workday/Dataverse data: "There's an open Senior Data Engineer role. I've prepared a draft requisition — want me to kick off the hiring workflow?"
2. **Hiring Manager** reviews and approves. PA triggers workflow via Durable Functions, acting OBO the manager.
3. **Agentic loop** loads budget + job-design skills, calls Workday/Dataverse MCP (routed through APIM AI Gateway), writes results to state store, exits
4. **Durable Functions**: budget needs Finance BP approval → `wait_for_external_event` (zero compute)
5. **Finance BP's PA** receives the approval request, presents it in Teams or via Adaptive Card with context and recommendation. Finance BP reviews and approves.
6. **PA** processes the decision, writes to state → `raise_event` → Durable Functions resumes next phase
7. **HR BP's PA** (Fleet Manager) monitors all workflows, proactively surfaces: "2 of your 20 workflows need attention. Workflow-456 has a compliance flag — here's what happened and my recommendation."
8. **HR BP** acts on the recommendation via Control Plane UI or directly in Teams. The other 18 workflows run autonomously.

3. Fleet Manager Agents

Domain-scoped GHCP SDK Hosted Agents that govern other agents. WPP cannot manage fleet-scale agentic operations relying on human eyes alone — agentic governance is required.

Multiple instances by domain: hiring fleet manager, finance fleet manager, compliance fleet manager. Each scoped to its domain's telemetry and workflows. If one degrades, its domain falls back to unfiltered view while others continue.

Inputs: telemetry events from agentic loops and Durable Functions via Event Grid. Enforcement events from Foundry Guardrails and APIM.

Outputs: fleet health assessment, prioritised exception queue, contextual recommendations, compliance alerts — pushed to Control Plane UI via SignalR.

What the operator sees: Fleet Manager's assessment. Exception-only view — only flagged workflows are visible. For each: what happened, why it stopped, recommended action, options. Bulk approval queue for batched low-risk decisions. Autonomy dials for per-workflow thresholds.

Delivers WPP's Refs 31.1-31.5: exception-only surfacing, situational awareness, skill amplification, AI-driven prioritisation.

4. Workflow Durability

Durable Functions handles: event routing, phase boundaries, HITL waits (zero compute, days/weeks), timer escalation, parallel coordination, checkpoint/replay, geo-replicated state.

Durable Functions does not handle: task reasoning, tool calls, skill execution, model selection, human interaction — all handled by agentic loops.

Workflow state store ([Cosmos DB](#) / Dataverse): phase state, context, candidate data, approval records, append-only action ledger (revocable/non-revocable tracking), OTEL span summaries. Cosmos DB provides [multi-region writes with automatic failover](#) and [continuous backup with point-in-time restore](#).

HITL pattern: agentic loop determines human input needed → composes contextual message/Adaptive Card → sends to appropriate human via their preferred surface → loop exits. Human responds → response triggers new agentic loop that processes the decision → `raise_event` → Durable Functions resumes next phase. Bulk approval: raise events on multiple instances simultaneously.

5. Observability

Layer	Source	Purpose
Foundry Tracing (OTEL)	Every agentic loop via GHCP SDK OTEL TracerProvider	Raw telemetry: model calls, tool calls, tokens, latency, cost. Drill-down for situational awareness. Foundry Agent Monitoring Dashboard out-of-box.
APIM AI Gateway metrics	Token usage , latency, errors per model/tool/agent	Central cost tracking and rate limit monitoring across all AI assets.
Fleet Manager assessment	Event-driven reasoning over telemetry	Intelligent layer: fleet health, anomaly detection, SLA risk, exception prioritisation. Default Control Plane view.

OTEL spans carry workflow ID, phase, jurisdiction, model, token count. Cost attribution per workflow, phase, model, and consumer visible in Control Plane and Foundry dashboards.

6. Compliance & Governance — Three Enforcement Layers

Compliance is enforced externally. The LLM cannot bypass these.

Layer	Technology	Status	Enforcement
Foundry Guardrails	Tool call/response interception, PII detection, Task Adherence	Preview	Intercepts tool calls before execution. Blocks PII in outputs. Detects policy drift from system message. Four intervention points : input, output, tool-call (preview), tool-response (preview).
APIM AI Gateway	Model routing, content safety, token limits	GA	Jurisdiction-based model routing (DE workflow → EU endpoint only). <code>11m-content-safety</code> policy. Token rate limiting. Semantic caching. All MCP tool calls governed (auth, rate limits, content filtering).
Agent 365 + Entra	Identity, access control, DLP, threat detection	GA May 2026	Per-agent tool access via RBAC on downstream resources. Conditional Access policies on agent identities . Purview DLP on agent interactions . Defender for threat detection .

Policy-as-code: APIM policies version-controlled via APIOps CI/CD. Foundry Guardrails configured per agent. Jurisdiction-specific skills define compliance rules declaratively.

Jurisdiction switching: workflow state carries `jurisdiction`. APIM routes to region-appropriate model endpoint. Foundry Guardrails enforce PII and content safety universally. Jurisdiction-specific skills (right-to-work,

works council, GDPR consent) load automatically based on workflow context. Task Adherence guardrail catches drift from jurisdiction policy in the system message.

Audit: [Azure Log Analytics](#) with archive tier (7-12yr retention; immutability enforced via Azure Storage export with immutability policies). Every tool call, model call, enforcement decision, and human interaction logged. Queryable via Log Analytics and [Microsoft Sentinel](#).

7. Builder Experience

Requirement	Approach
Pro-code SDK (Must)	GHCP SDK Python (primary) + TypeScript for skills and MCP servers
Low-code visual builder (Must)	Copilot Studio: visual workflow designer. Supported via Agent 365 (not Foundry Hosted Agents). Custom Control Plane UI natively supports Copilot Studio agents alongside GHCP SDK agents. Not recommended for complex autonomous workflows — available for citizen developers building simpler agent experiences.
Low-code configuration (Must)	Control Plane UI: skill library (backed by Azure API Center), tool catalog (APIM), governance editor, autonomy dials, template fork-and-customise
Agentic builder (Could)	Agentic loop generates new SKILL.md files from natural language. Registered in API Center. Human reviews and approves. This capability has been built and demonstrated.
Template library (Should)	Pre-built skill sets and workflow templates in Git. API Center integrates with GitHub Actions for syncing <code>skill.md</code> files from repos. Forkable and customisable.
Knowledge extraction (Could)	Threadlight accelerator: creates agentic loops with skills from interviews and unstructured data. Captures undocumented procedural knowledge from SMEs and produces machine-actionable SKILL.md files.

Skills registered in Azure API Center have lifecycle management (Design → Preview → Production → Deprecated), declared allowed tools, and GitHub Actions integration for `skill.md` files.

8. Control Plane — Two Layers

Foundry Control Plane (platform governance)

Foundry Control Plane provides the platform-level management layer. It is an existing product, not custom build:

- Fleet health dashboards: active agents, run completion rates, compliance posture, cost efficiency, prohibited behaviours
- Agent inventory across all platforms (Foundry-native, custom, external) with lifecycle operations
- Model registry with quota management
- Tool registry with APIM governance
- Continuous evaluation and monitoring with deep links for debugging
- Compliance enforcement (guardrails configuration)
- Integration with Defender (threat detection) and Purview (DLP)

This covers WPP's requirements for agent registry (8.5), model registry (8.1), observability (8.14), agent lifecycle (22.3), and compliance posture monitoring.

Custom Control Plane UI (operator experience)

Foundry Control Plane monitors the platform. WPP's requirements go further — they need an operator experience where 1 human governs 20-50 concurrent workflows with intelligent exception surfacing. This requires a custom React application powered by the Fleet Manager agent:

Capability	Why custom	WPP Refs
Fleet Dashboard (Fleet Manager's assessment)	Foundry shows platform health. WPP needs workflow-level status, SLA tracking, agency/market/jurisdiction filtering. Fleet Manager reasons about what to show.	31.1
Exception-Only Queue	Foundry does not filter to "the 2% needing attention." Fleet Manager composes this intelligently based on business impact, confidence, SLA urgency.	31.2
Instant Situational Awareness	Click any workflow → what agents did, what stopped progress, recommendations, options. Requires OTEL drill-down + Fleet Manager's contextual reasoning. <5 seconds.	31.3
Bulk HITL	Batch similar decisions (e.g., 8 interview schedules), approve in one action. Raises events on all waiting Durable Functions instances.	31.4
Autonomy Dials	Per-workflow threshold sliders (auto-shortlist %, HITL gates). Writes to config store, effective immediately.	21.1
Skill Amplification	Fleet Manager proactively surfaces relevant policy, precedents, recommended approach when operator is uncertain.	31.5
Role-Based Operator Views	HR BP sees hiring workflows. Finance BP sees budget gates. IT Ops sees provisioning. RBAC-filtered.	10.1

Capability	Why custom	WPP Refs
Cost Dashboard	Per-workflow cost attribution from OTEL + APIM token metrics.	26.4

The custom UI consumes data from: **Application Insights APIs** (agent run traces, token usage, cost, error rates), **Foundry REST APIs** (agent inventory, model deployments), **APIM metrics** (token consumption, rate limit status per model/tool), **workflow state store** (Cosmos DB / Dataverse — phase status, approvals, action ledger), and **Fleet Manager agent assessments** (pushed via SignalR).

9. Voice, Video, Avatar

These are MCP tools and skills loaded by agentic loops when needed.

Capability	Approach
Voice screening	GPT-Realtime as speech-to-speech front end. ACS Call Automation for telephony. MAI-Transcribe-1 for transcription. MAI-Voice-1 for TTS (preview). Tool calls to GHCP SDK backend for reasoning and scoring.
Video meeting notes	Teams bot joins meeting (Bot Framework, GA). Post-meeting transcript via Graph API MCP. Agentic loop processes into structured notes and scores. Real-time in-meeting speaking achievable via ACS integration but primary value is intelligent note-taking.
Avatar onboarding	Agentic loop generates personalised script. HeyGen API MCP produces branded video. Avatars configurable (appearance, voice, branding) and persistent across sessions.

10. Component Summary

Component	Technology	Role
Agentic Runtime	GHCP SDK (MIT, Python/TypeScript)	Autonomous reasoning, skills , MCP tools , hooks
Hosting	Foundry Hosted Agents	Containerised deployment, scaling. GHCP SDK wrapped in container exposing Responses API (custom adapter).
Workflow Durability	Azure Durable Functions (GA)	Event routing, HITL waits, phase coordination

Component	Technology	Role
Fleet Governance	Fleet Manager Agent (GHCP SDK)	Agentic monitoring, exception composition
Central AI Gateway	APIM AI Gateway (GA)	Govern models, tools, agents, skills centrally
Control Plane	Foundry Control Plane (GA)	Agent fleet management, evaluation, compliance
Agent Lifecycle	Agent 365 + Entra Agent ID	Identity, RBAC. GA May 2026 — integration with Hosted Agents needs POC validation. Supports Copilot Studio agents natively.
Control Plane UI	Custom React	Fleet dashboard, HITL queue, autonomy dials
Observability	Foundry Tracing (OTEL) + GHCP SDK OTEL	Telemetry, cost attribution
Evaluation	Foundry Evaluators	Task adherence, quality, safety, continuous evaluation
Compliance Layer 1	Foundry Guardrails	Tool call/output interception, PII, task adherence
Compliance Layer 2	APIM AI Gateway	Model routing, content safety, token limits
Compliance Layer 3	Agent 365 + Entra	Identity, access control, DLP, threat detection
Human Surfaces	M365 Agents SDK + Custom UI	Copilot (Teams), Email (Adaptive Cards), Control Plane, Web, Voice
Integration	MCP Servers (via APIM)	Workday, Greenhouse, LinkedIn, ServiceNow, Graph, Dataverse, ACS, HeyGen
Skills	SKILL.md + Azure API Center	Declarative, composable, lifecycle-managed, GitHub Actions integration
Voice	GPT-Realtime + ACS	Speech-to-speech with agentic backend
Audit	Azure Log Analytics	7-12yr retention (immutability via Azure Storage export)
Knowledge Extraction	Threadlight accelerator	Interview-based knowledge capture into SKILL.md files

11. POC 1: Finance Procure-to-Pay

Scope: 30-50 concurrent invoice workflows managed by a Finance Controller via the Control Plane.

Skills: invoice parsing (OCR via Azure Document Intelligence), three-way match (PO matching), GL coding and cost centre allocation, payment file generation, statement reconciliation.

MCP integrations: Workday, Dynamics 365 F&O, Maconomy — all governed through APIM AI Gateway.

Demonstrates: multi-phase workflow orchestration via Durable Functions, HITL approval gates (Finance BP interacts via Adaptive Card in Outlook, routed through agentic loop), bulk approval for batched low-risk items, rollback/compensating actions, Fleet Manager monitoring 30-50 concurrent workflows, exception-only Control Plane view, OTEL cost attribution per invoice, Foundry Guardrails (PII detection, content safety).

12. POC 2: HR Talent Lifecycle

Scope: 15-20 concurrent hiring workflows managed by an HR Business Partner via the Control Plane. Five human participants across four timezones.

Skills: CV parsing and candidate scoring, screening questionnaires, voice screening (GPT-Realtime + ACS), job design and comp benchmarking, right-to-work checks, GDPR consent management, EU AI Act classification, offer letter generation, JML onboarding workflow, avatar welcome video (HeyGen).

MCP integrations: Greenhouse ATS, LinkedIn Recruiter, Workday (hiring), Microsoft Graph (calendar/email), ServiceNow (IT provisioning), Azure Communication Services (voice), HeyGen (avatar) — all governed through APIM AI Gateway.

Human surfaces: Hiring Manager via M365 Copilot in Teams (agent surfaced via M365 Agents SDK), Finance BP via email Adaptive Cards, candidate via web portal + voice, IT Ops via ServiceNow, HR BP via Control Plane.

Demonstrates: all POC 1 capabilities plus — voice screening with structured scoring, CV parsing with crystallisation pipeline (skill promotion from generative to deterministic, managed in API Center), episodic memory from workflow state store (recall past hires levelled too low), A2A interop with external candidate agent (governed via APIM), jurisdiction-aware compliance (USA vs Germany enforcement switching via APIM routing + jurisdiction-specific skills + Foundry Guardrails), autonomy dials (configurable auto-shortlist thresholds), skill amplification (Fleet Manager surfaces policy + precedents), process evolution (Fleet Manager proposes improvements after completed workflows), synthetic CV evaluation (500 CVs via Foundry Evaluators), avatar onboarding video, **Threadlight** knowledge extraction demo (interview HR SME, produce executable skills).

13. Known Constraints

Constraint	Impact	Mitigation
GHCP SDK in tech preview	API surface may change	GitHub Copilot is GA. SDK stabilising. Core patterns (skills , MCP , hooks) proven in production. MIT open-source.
Foundry Hosted Agents: max 5 replicas per deployment (preview)	Scaling ceiling	Multiple deployments, or Azure Container Apps with Foundry telemetry. Preview limit expected to increase at GA.
Foundry Guardrails tool call interception (preview)	May not be GA for POC	GHCP SDK session hooks provide equivalent enforcement at code level. Guardrails are additive.
APIM A2A agent governance (preview)	A2A features still maturing. A2A is not required for core architecture — only for POC2 external candidate agent demo.	HTTP gateway primitives work today. Purpose-built A2A policies emerging.
Skills in Azure API Center (preview)	Skill registry is new. API Center does not have native Git sync — uses GitHub Actions workflows.	Core skill execution is GHCP SDK native. API Center adds governance layer.
GHCP SDK + Foundry Hosted Agents integration not documented	Hosting adapter needs custom work	Hosted Agents accept any container image that exposes the Responses API protocol. The adapter must translate between Responses API and GHCP SDK session management. This is the primary integration engineering task for the POC.
Agent 365 GA: May 2026	Not yet GA. Integration with Foundry Hosted Agents unclear.	In preview. Whether Hosted Agents auto-register in Agent 365 or require manual onboarding needs validation. Entra Agent ID (the identity layer) is usable independently of Agent 365. Agent 365 natively supports Copilot Studio agents; our Control Plane will also support them.
MAI-Voice-1	In public preview, not GA. No SLA.	Screening calls use GPT-Realtime (GA) as primary. MAI-Voice-1 is additive for TTS quality.
Copilot Studio on Foundry Hosted	Copilot Studio is not supported on Foundry	Copilot Studio agents supported via Agent 365. Custom Control Plane UI natively supports both

Constraint	Impact	Mitigation
Agents	Hosted Agents.	Copilot Studio and GHCP SDK agents.